

Standardizing Objective Measures of Presence in Immersive Virtual Environments

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ABSTRACT

Presence is the feeling of actually being located in a virtual environment, and has been subject to intensive research as long as virtual reality exists. Nowadays, it is commonly evaluated using subjective measures, which most of the time take the form of questionnaires. However, there have been doubts about whether measures of this type are able to truly capture the actual concept behind presence (if it exists, although this is highly likely). While working on my master's thesis, I found that it is quite difficult in practice to purposefully employ more objective measures into specific experimental designs. This is why I decided to dedicate myself to contributing ways to make this easier for as many scenarios as possible. My current work involves advanced pattern recognition and general methods from digital signal processing to analyze objective, continuous sensory data like eye-tracking and electromyography.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *User studies*.

KEYWORDS

presence, virtual reality, behavioral measures, physiological measures, breaks in presence, eye tracking, electromyography, pattern recognition

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1 INTRODUCTION

My name is Johannes Schirm, I am a PhD student at Nara Institute of Science and Technology in Japan. As a member of the Cybernetics and Reality Engineering (CARE) Laboratory, I am supervised by Professor Kiyoshi Kiyokawa, Associate Professor Nobuchika Sakata and Assistant Professor Naoya Isoyama. I am currently in the first year of my PhD studies and prepare to graduate in late 2023.

My first interaction with human-computer interaction (HCI) research was during my undergraduate studies, when I decided to

enter the most exciting placement position that I could find near my university. It included development of virtual reality applications for research purposes and work with virtual full-body avatars under the supervision of Betty Mohler [10] [9]. In a very different context, during my studies of game development, I found myself needing to evaluate general differences in user experience between two locomotion techniques in virtual reality. While reviewing more and more of the available research, I slowly discovered the enormous body of literature on presence. Often referred to as the sensation of “being there,” it has always been recognized as being one of the most important factors contributing to user experience, especially in virtual reality.

Although this notion is now slowly changing and a lot of progress has been made in strategically decomposing this multilayered concept in the last decade, I was delighted to see how relevant the topic still is and decided to invest my resources into an own contribution to this interesting field. This completely aligns with my personal interests, since I am currently most interested in becoming able to work scientifically and contribute my work in a sustainable, freely available way. There are many open research questions regarding presence, and although it is very challenging to gain generalizable insights, this topic inspires even more to get creative and find new approaches to understanding it better and putting it into practice.

2 RELATED WORK

Because presence is such a difficult-to-grasp concept and previous literature has spent a significant amount of effort alone on decomposing it into more concrete qualia, there is already a great variety of creative contributions available. Especially measures that rely on more objective data (in contrast to measures solely relying on subjective assessments of users) require a high degree of creativity with constructing the experimental setup, interpreting sensor data through generalizable and valid means, and integrating new insights into the body of literature without leaving room for misinterpretation. This section will summarize some of the most relevant literature for the presented topic, but makes no claim of reflecting the entire development of presence research. An excellent resource for this purpose is the literature survey by Skarbez et al., who have organized and reviewed all of the most relevant literature [14].

2.1 Concept

Perhaps one of the most difficult tasks in presence research has been the definition of the concept itself. For the longest time, presence was just defined as “being there,” but there have always been made efforts to find a more concise, but still generally valid definition. The in my opinion most comprehensive decomposition was

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presented by Slater, who broke the concept down into Place Illusion (PI) and Plausibility Illusion (Psi) [17]. If a virtual environment allows for high levels of immersion, PI—the feeling of “being there,” formerly referred to as presence—arises, and if a virtual environment allows for high levels of coherence (roughly equivalent to realism of events), Psi—the feeling that the virtual environment is real—arises. This correlation has also been summarized very well by Skarbez et al. [15] and will serve as the fundamental concept of presence for my own research.

2.2 Subjective Measures

Most of the time, subjective measures of presence take the form of post-study questionnaires and summarize user experience more generally after it has already ended. The Slater, Usoh and Steed questionnaire (SUS) [18], the Presence Questionnaire (PQ) by Witmer and Singer [20] and the Igroup Presence Questionnaire [12] are the most well-known instruments for this purpose. Bouchard et al. proposed a 1-item questionnaire [1], which users will be able to answer in real-time, but when viewing this instrument in the light of more recent literature, the caused break in presence might not outweigh the advantages gained by making measures in real-time.

Having users answer questionnaires about their experience is easy to implement and independent from the virtual environment—gathering objective data and interpreting it correctly is a very difficult task in comparison. However, Slater has already drawn attention to several problems that arise from solely relying on subjective measures to evaluate presence [16]. The recommendation by Skarbez et al. in their literature survey [14] to incorporate at least two different types of measures makes sense and is something I was interested in implementing for my master’s thesis.

2.3 Objective Measures

When working on my master’s thesis, I was immediately fascinated by the variety of physiological and behavioral measures of presence that researchers had already come up with. However, most of them depend on very specific features of the virtual environment they are applied in. The well-known virtual cliff experiment by Meehan et al. [6], for example, showed how the difference in heart rate (a physiological measure) correlates to subjective presence when users are confronted with a seemingly dangerous cliff in the virtual environment. While this is one of the clearest cases available in the literature, its practical uses are limited to virtual environments containing this kind of stressful situation. This might be an important reason why in practice, most studies eventually end up relying on subjective measures of presence.

2.3.1 Physiological Measures. The two most common physiological measures are difference in heart rate and skin conductance, a prominent example for the former being the virtual cliff experiment [6]. One example of a study which incorporated both of these measures is the work by Deniaud et al. [2]. The authors came to the conclusion that “presence measurement can’t be only based on subjective measures.” There have also been attempts to utilize technologies like Electroencephalography (EEG) [4] and Functional magnetic resonance imaging (fMRI) [3], where the authors had to come up with creative solutions. However, interpreting raw sensor

data like this in terms of presence is difficult and currently not as far advanced as it would be necessary to use it in studies.

2.3.2 Behavioral Measures. Behavioral measures are mostly transparent to users, but they heavily depend on the specific circumstances they are applied in. However, these kinds of measures are very experimental and probably the most creative ways to measure presence. The experimental design of my master’s thesis was very much inspired by the work of Nichols et al. and their early studies of startle responses in virtual reality [7]. Sheridan already introduced the ideas for behavioral measures of presence very early [13, p. 4], which for example included the analysis of social responses of users with regards to naturalness. One example of a behavioral measure completely independent of the virtual environment is the comeback rate by Thie and van Wijk [19].

3 RESEARCH QUESTIONS

One problem with almost all currently available objective measures is that there is no standardized way one can implement them into own experimental designs. In order to collect as much relevant data as possible, the final study for my master’s thesis therefore included several different measures, all slightly adjusted to the specific use case. This experience motivated me to find ways to offer other researchers and students in a situation similar to mine a more general way to objectively evaluate presence.

The following research questions are most relevant to this:

- Which “interfaces” to the user could one make use of to reliably infer certain information about their level of presence—without intruding into their experience?
- What are behavioral patterns that are widely present in users of most types of virtual environments?
- And how could one construct a generalizable real-time measure that would, independently from the many individual variances in users, reliably detect those patterns?

As time passes, more and more (sensor) data about the user will become accessible by default. One example for this is eye-tracking, which is already embedded in virtual reality headsets like the HTC Vive Pro Eye. I believe that the ideal way to make use of this data in terms of evaluation of presence is to maintain a free and open-source, cross-platform “toolkit” which removes many of the hurdles of adding objective measures of presence into the average experimental design.

4 METHODOLOGY

One very interesting and in terms of methodology for my research certainly relevant paper is the work by Liebold et al. [5], who were able to use a high-quality video game to investigate breaks in presence. They collected data from a wide variety of interfaces and creatively thought of many different ways to intentionally break the presence of users. This is very close to my current research plan, which was considerably shaped through conversations with Richard Skarbez and the members of our laboratory. The basic idea is to provide users with a compelling virtual environment and a basic, rather repetitive task to accomplish, collect behavioral and sensory data from more than one objective data source, and intentionally induce multiple breaks in presence. By utilizing machine learning

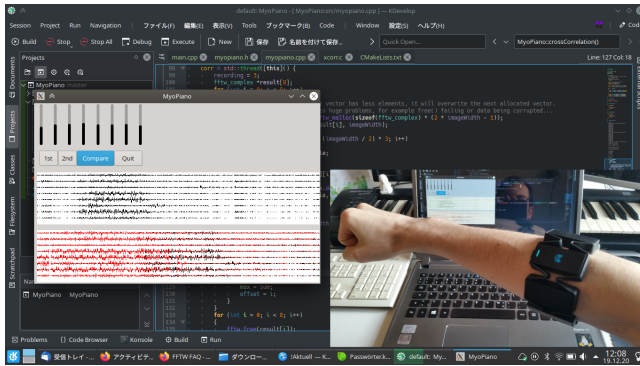


Figure 1: The Myo armband, an electromyographic user interface device, connected to my test application. Electrical pulses in my muscles are being measured at eight points and sent to the computer at a rate of around 200 Hz. For testing, the cross-correlation between two recorded signals is shown in red on top of the second signal. It represents their similarity for different offsets of the first signal.

and general tools from digital signal processing, I hope to be able to pinpoint common patterns in the recorded data shortly after a break in presence was induced and compare these to appropriate variants which were recorded in sections of the user’s experience that provided ideal conditions for very high levels of presence.

When selecting the interfaces to collect data from, I now think that those with continuous data will provide the most value. With continuous data, I mean that the data is available for almost every interaction with the virtual environment, and not only for specific occasions like standing at a virtual cliff or being startled by a ghost. Instruments like skin conductance, startle reflexes and difference in heart rate unfortunately do not provide continuous data and are therefore less fit for this purpose. This is why I plan to focus on more basic, but continuously collectable instruments like eye-tracking and electromyography (EMG, like EEG, just with the nerve impulses going through muscles), which can also be collected without intruding too much into the user’s experience.

The latter is something that I am exploring just at the time of this writing. Using an armband like the Myo (see Figure 1), EMG signals can be collected very easily, without the need to build a custom device out of single electrodes. A comparison of different EMG sensors by Pizzolato et al. [8] has shown that even consumer hardware like the Myo armband can yield accurate results. Eye-tracking is already integrated into virtual reality headsets like the HTC Vive Pro Eye and can be incorporated into the experimental design very easily. I am looking forward to learning more about the technical processes needed to collect this data, but also about contemporary tools that make more general analyses like pattern recognition possible.

5 RESULTS

Even though my PhD project is still in its beginning stages, I have already been able to publish first work on this very topic in the context of my master’s thesis with the support of Gabriela Tullius

and Jacob Habgood [11]. We were not able to observe significant differences between the two compared locomotion techniques in virtual reality. But much more interestingly, we were able to suggest that the extent to which users featured startle reflexes when they encountered an unexpected ghost behind them bear relation to their subjective presence. Since there is no recent literature exploring this type of behavioral measure, I originally planned to center my PhD around investigating it more deeply and finding ways to generalize it, so it could be used in future experimental designs. But while reviewing more of the recent literature and having conversations with professionals from the field, it became clear that startle reflexes might be very difficult to generalize and it might make more sense to primarily focus on continuous sensory data.

6 OUTLOOK

After having received advice on what aspects to primarily focus on, I plan to start implementing the first user-study. In parallel, I plan to investigate different instruments of data collection, especially eye-tracking and electromyography. For both of them, I would like to implement an experimental prototype which will not necessarily be thematically related to my final research question about presence. For example, in order to make progress quickly, I plan to experiment with different means to recognize patterns in electromyographic data produced by playing the piano. Different to conventional gesture recognition (of one specific configuration of muscles receiving electrical activation pulses), this will provide me with a more individual and easily recognizable temporal pattern.

Regarding eye-tracking, I am very interested in reading speed with different languages. When reading a text that I have never read before, I would expect that a paragraph of German (which I have native abilities in) would yield very different general eye movement patterns than a text in Japanese (which I am currently in the middle of learning). Differentiating German from English (which should be similar to German to me, although I feel that I am not as fluent in it) should be a much more difficult task, which would allow me to gain insights into how accurately different methods are able to distinguish these kinds of eye-tracking patterns. Again, this is not directly related to presence, but I plan to review some of the existing literature on eye-tracking analysis in the context of reading speed and get a deeper understanding of the science behind it before applying it to the complex case of presence.

I think that the above mentioned tasks help me to prepare the first user-study as well as possible. My long-term goal is to then start developing a free and open-source, cross-platform “toolkit” which incorporates the insights I was able to gain while experimenting with various technologies and analyzing the results of the user-study. Having a network-based (or generally cross-platform) solution is crucial in my opinion, since virtual environments tend to be implemented in many ways, and it will be difficult to provide the same functionality everywhere. My idea for how to implement this is to keep all the logic and user-interfaces cross-platform, and to provide very simple adapter plug-ins for the various game engines like Godot, Unreal or Unity. My hope is that this contribution will provide a basis for maintaining an easier and freely accessible way to allow researchers and students to have more confidence in their conclusions on presence measurements.

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